Bell

Parametrisation

The Bell distribution is

$$\text{Prob}(y) = \frac{\lambda^y \exp(1 - \exp(\lambda))B_y}{y!}$$

for responses $y = 0, 1, 2, \ldots$, where B_y are the Bell-numbers ($B_2 = 2, \ldots, B_5 = 52, \ldots, B_8 = 4140$, etc). The moments are $\text{E}(y) = \lambda \exp(\lambda)$ and the variance is $\text{Var}(y) = \lambda \exp(\lambda)(1 + \lambda)$.

Link-function

The mean is linked to the linear predictor by

$$\lambda \exp(\lambda) = E \exp(\eta)$$

where $E > 0$ is a known constant which defaults to 1.

Hyperparameters

None.

Specification

- `family="bell"`
- Required arguments: (integer-valued) y and E (default 1).

Example

This example estimate parameters from simulated data.

```
library(VGAM) ## dbell
library(gsl)  ## lambert_W0

dbell <- function(y, theta)
    return (theta^y * exp(1-exp(theta)) * bell(y) / factorial(y))

pbell <- function(y, theta)
    return (sum(dbell(0:y, theta)))

rbell <- function(n, theta) {
    ## brute-force in lack of anything easy available
    stopifnot(length(theta) == 1)
    ymax <- 0
    cdf <- 0
    while(cdf < 0.99999) {
        ymax <- ymax + 10
        cdf <- pbell(ymax, theta)
    }
    y <- 0:ymax
    prob <- dbell(y, theta)
    return (sample(y, n, prob = prob, replace = TRUE))
}

## theta <- 2
## hist(rbell(1000, theta))
```

```
n <- 300
x <- rnorm(n)
eta <- 1 + 0.1 * x
mu <- exp(eta)
y <- numeric(n)
for(i in 1:n) {
    theta <- lambert_W0(mu[i])
    y[i] <- rbell(1, theta)
}

r <- inla(y ~ 1 + x, data = data.frame(y, x), family = "bell")
summary(r)
```

Notes

None.